

Contradiction and Cases

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Review

To prove $P \Rightarrow Q$ directly, assume P and derive Q .

To prove it by contraposition, assume $\neg Q$ and derive $\neg P$.

Both techniques prove an implication by following a valid propositional form.

We now consider proof by contradiction and proof by cases.

Proof by contradiction

To prove a proposition P by contradiction:

1. Assume $\neg P$.
2. Use this assumption to derive a proposition R and its negation $\neg R$.
3. Conclude that the assumption $\neg P$ is impossible, so P is true.

The form $R \wedge \neg R$ is a contradiction: it is false under every truth assignment.

Why contradiction works

Assume $\neg P$ and derive $R \wedge \neg R$ by valid reasoning.
This proves that the following implication is true:

$$\neg P \Rightarrow (R \wedge \neg R).$$

The conclusion $R \wedge \neg R$ is always false.

A true implication with a false conclusion must have a false hypothesis:
a true hypothesis and a false conclusion would make it false.

Therefore $\neg P$ is false, so P is true.

Rational and irrational numbers

A real number x is **rational** if

$$x = \frac{a}{b}$$

for some integers a and b with $b \neq 0$.

A real number that is not rational is **irrational**.

Two positive integers are **coprime** if their only common positive divisor is 1.

Every positive rational number can be written as a/b with positive, coprime integers a and b .

The square root of two

Theorem

The number $\sqrt{2}$ is irrational.

Assume for contradiction that $\sqrt{2}$ is rational.

Then

$$\sqrt{2} = \frac{a}{b}$$

for some positive, coprime integers a and b .

Squaring both sides gives

$$a^2 = 2b^2.$$

Both integers are even

From $a^2 = 2b^2$, the integer a^2 is even.

Since an integer with an even square is even, $a = 2k$ for some integer k .

Substituting gives

$$2b^2 = (2k)^2 = 4k^2,$$

so

$$b^2 = 2k^2.$$

Thus b^2 is even, and therefore b is even.

The contradiction

We have shown that both a and b are even.

Therefore 2 is a common positive divisor of a and b .

This contradicts the choice of a and b as coprime integers.

Hence the assumption that $\sqrt{2}$ is rational is false.

Therefore $\sqrt{2}$ is irrational.

Proof by cases

A **proof by cases** divides the possible situations into cases and proves the conclusion in every case.

The cases must be **exhaustive**: together they must cover every possibility.

The cases need not be disjoint, although disjoint cases often make a proof clearer.

Unlike a brute-force proof by exhaustion, a case proof can use a small number of broad cases, each containing many individual possibilities.

Selecting socks

A drawer contains one pair of gray socks, one pair of white socks, and one pair of black socks.

Claim

Any selection of four socks contains two socks of the same color.

Consider the first three selected socks. There are two cases:

1. two of them already have the same color;
2. all three have different colors.

These cases cover every possible selection of the first three socks.

The two cases

Case 1: Two of the first three socks have the same color.

Then the required pair has already been selected.

Case 2: The first three socks have different colors.

There is one gray, one white, and one black sock among them. The fourth sock has one of these three colors, so it matches one of the first three.

The claim holds in both exhaustive cases; therefore it holds for every selection of four socks.

Cases in propositional logic

Suppose $P_1 \vee P_2 \vee \cdots \vee P_n$ is true and each case implies Q :

$$P_1 \Rightarrow Q, \quad P_2 \Rightarrow Q, \quad \dots, \quad P_n \Rightarrow Q.$$

Then Q is true, because at least one case P_i must be true.

A common two-case split is

$$P \quad \text{or} \quad \neg P.$$

The form $P \vee \neg P$ is a tautology, so these two cases are exhaustive.

A parity proof

Theorem

For every integer n , the integer $n^2 + n$ is even.

Every integer is either even or odd, so these form two exhaustive cases.

Case 1: If $n = 2k$, then

$$n^2 + n = 4k^2 + 2k = 2(2k^2 + k).$$

Case 2: If $n = 2k + 1$, then

$$n^2 + n = 4k^2 + 6k + 2 = 2(2k^2 + 3k + 1).$$

In both cases the expression is twice an integer, so it is even.

Choosing a proof technique

structure of the problem	useful technique
hypothesis gives usable algebra or definitions	direct proof
negated conclusion is easier to use	contraposition
negating the claim leads to impossibility	contradiction
a small set of broad possibilities	cases

A theorem may admit several correct proofs; the best technique is the one that makes the reasoning easiest to verify.

Summary

- ▶ A contradiction proof assumes the negation of the claim and derives an impossibility.
- ▶ The proof that $\sqrt{2}$ is irrational contradicts the existence of a reduced fraction for $\sqrt{2}$.
- ▶ A proof by cases establishes the conclusion separately in exhaustive cases.
- ▶ Cases may be broad and need not list every individual possibility.